

REAVER — Rendezvous & Proximity Operations (RPO) Analysis

Close-range ΔV budget and AOCS sizing — debris capture and Recycling-Hub handover

1. Introduction

The REAVER mothership (MS) performs close-range RPO twice per debris: once to **capture** a tumbling debris object, and once per cycle to **hand it over** at the cooperative Recycling Hub (RH). These maneuvers set the close-range ΔV budget — the long-range orbit transfers are sized separately by the mission optimizer — and they drive the AOCS sizing.

This report walks through the two campaigns — **Part 1: Debris RPO** and **Part 2: RH docking cycle** — using four trajectory figures. Each maneuver is a sequence of short translations between hold points, plus the attitude work (spin-up, momentum dump, re-point). The figures are the primary explanation aid; the tables give the per-phase numbers and the equations close the loop. All trajectories use the target **Hill (LVLH) frame**: R-bar (radial), V-bar (along-track), H-bar (cross-track).

2. Assumptions

#	Assumption
1	Relative motion is the linearised Clohessy-Wiltshire model about a circular GEO orbit, mean motion $n = \sqrt{\mu/a^3} \approx 7.11 \times 10^{-5}$ rad/s.
2	Each translation is a rest-to-rest two-impulse CW transfer ; transfer time keeps the mean speed $\approx \frac{1}{2} \cdot V_{\text{MAX}}$, with $V_{\text{MAX}} = 0.05$ m/s (low-impact docking limit).
3	Burns are impulsive ; propellant follows Tsiolkovsky with MS monopropellant $I_{\text{sp}} = 253$ s $\rightarrow v_e = 2481$ m/s.
4	Attitude uses 24 RCS thrusters (4 N each); the slew/detumble couple is $\tau = 2 \cdot (3 \cdot 4 \text{ N}) \cdot 1.05 \text{ m} = 25.2$ N·m. Momentum-dump propellant = $H/(\text{arm} \cdot v_e)$ is thruster-count independent.
5	Reference debris is COMSATBW-1 (2440 kg, 17.2 m span) tumbling at 6 °/s (1 rpm) — the REQ-MIS-06 ceiling and AOCS-dimensioning case.
6	Inertia tensors are box models : MS $3.0 \times 3.0 \times 3.5$ m, tug $0.75 \times 0.75 \times 1.2$ m, debris bus $2.5 \times 2.5 \times 3.0$ m + panels.
7	Keep-out spheres (KOS) are centred on the target: $\text{KOS1} = (\text{panel span}/2) \cdot 1.20$; KOS2 from hardware geometry (see §5).
8	Slew rates are bounded by V_{MAX} at the payload tip ($\omega \approx V_{\text{MAX}}/r$): 0.5 °/s for the dock re-point, 0.2 °/s for the clamp settle. Costed as a single momentum build.
9	The RH is cooperative (known state, fixed docking axis): no spin-synchronisation, only a ~ 30 min go/no-go dwell.
10	A +50 % abort margin is applied to the nominal debris-RPO ΔV (not to the detumble line).
11	TBD (flagged) : robotic-arm length (5.0 m placeholder), RH structural size, and debris panel orientation during the RH dock.

3. RPO Phases

Part 1 — Debris RPO (Fig 1)

The debris is **uncooperative** — silent, tumbling, no fixed docking axis — so the MS cannot just pick a corridor. It first **observes from a stable hold**, then approaches along the tumble axis it has measured. The hold is on **V-bar**, the only naturally fixed CW hold point (a body at rest at an along-track offset stays put; an H-bar offset would oscillate back through the plane). The capture corridor is then flown along the **identified tumble axis** (modelled as H-bar).

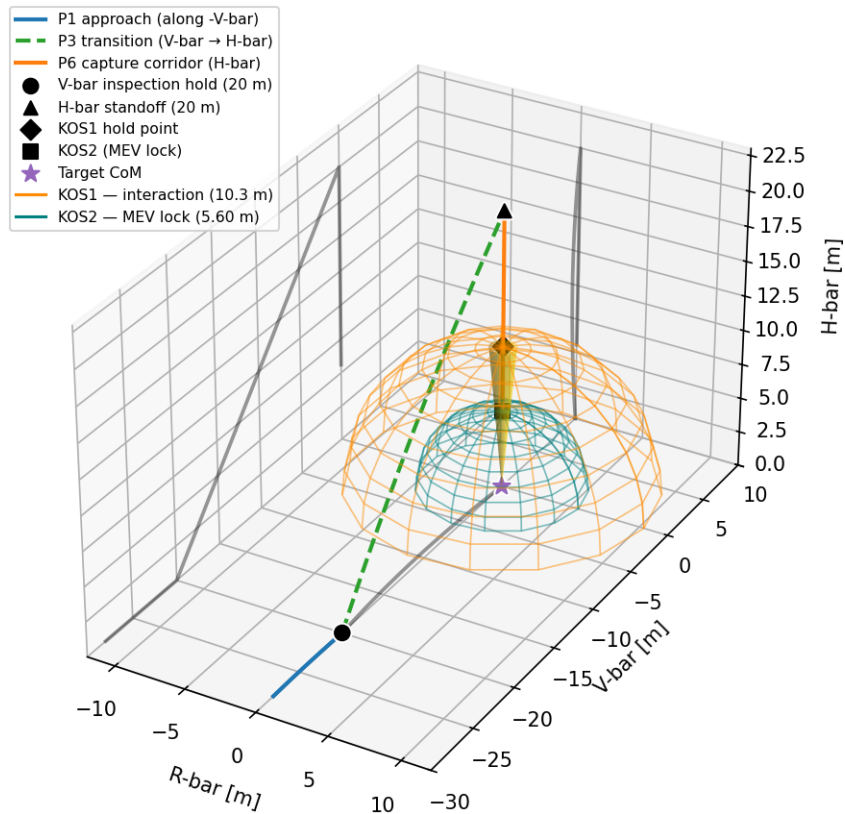


Fig 1. Two-stage debris RPO: (blue) approach along $-V\text{-bar}$ to the $V\text{-bar}$ inspection hold (20 m); (green) transition around the keep-out sphere onto the tumble axis; (orange) capture corridor down the tumble axis through KOS1 (10.3 m) $\rightarrow$ KOS2 (5.6 m, MEV lock), inside the 5° cone.

- **P1 Approach:** cut-off (150 m) $\rightarrow$ 20 m **V-bar** inspection hold.
- **P2 Inspection:** 8 h dwell on the stable $V\text{-bar}$ hold — measure spin rate + tumble axis, ground go/no-go (you must know the axis before approaching it).
- **P3 Transition:** reposition around the keep-out sphere onto the identified tumble axis ($H\text{-bar}$), then close to KOS1.
- **P4 Arm extension:** deploy the arm + tug at KOS1 (before spin-up).
- **P5 Spin-sync:** spin the deployed assembly up to match the tumble (spin-up + inertia growth sum to the same total either way).
- **P6 MEV lock + detumble:** final KOS1 $\rightarrow$ KOS2 approach, then the combined-body momentum dump — the largest ΔV line and the AOCS sizing case.
- **P7 Retreat:** MS backs off to a safe standoff after tug release.

Part 2, Phase A — Tug Meet (Fig 2a)

At the RH the MS first **meets the cooperative tug+debris** and clamps it with the bare robotic arm. Being cooperative, the approach is a direct **V-bar** corridor — no $H\text{-bar}$ detour, no spin-sync.

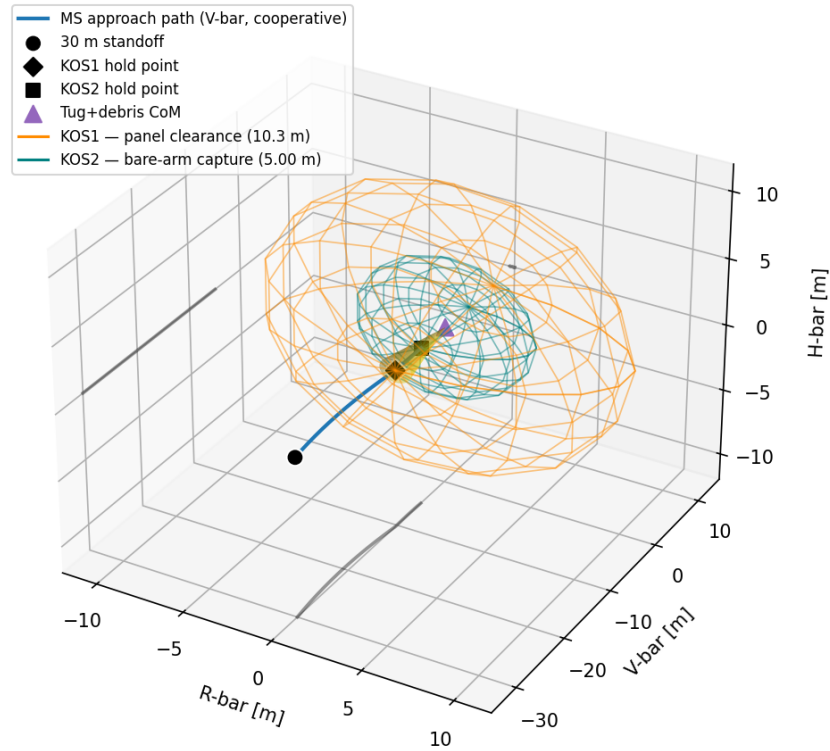


Fig 2a. Tug-meet close approach: 30 m standoff → KOS1 (10.3 m) → KOS2 (5.0 m, **bare arm length** — no tug on the MS arm yet, so tug/2 is not added).

Part 2, Phase B — RH Dock (Fig 2c then Fig 2b)

The MS now carries the combined body (MS + **extended arm** + **tug** + **debris**) back to the RH and hard-docks via a dedicated port on **top** of the bus — the arm stays extended, holding the payload to the side. Split into a far-range carry and a close approach.

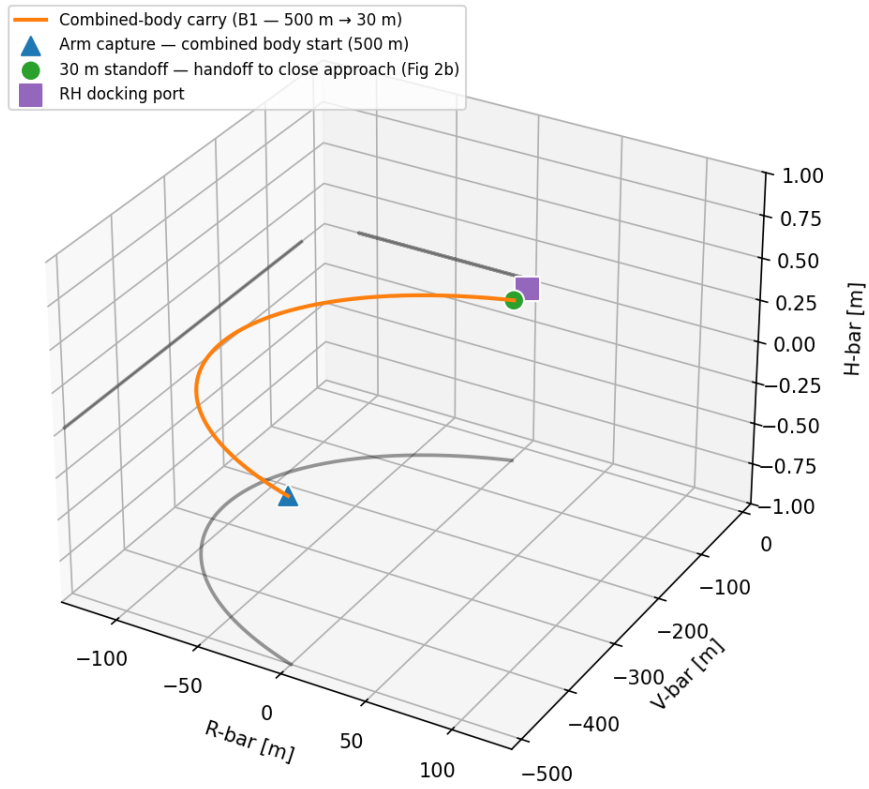


Fig 2c. Far-range carry (B1): combined body from the 500 m capture hold to the 30 m standoff. The R-bar excursion is the natural CW coasting arc; green = handoff.

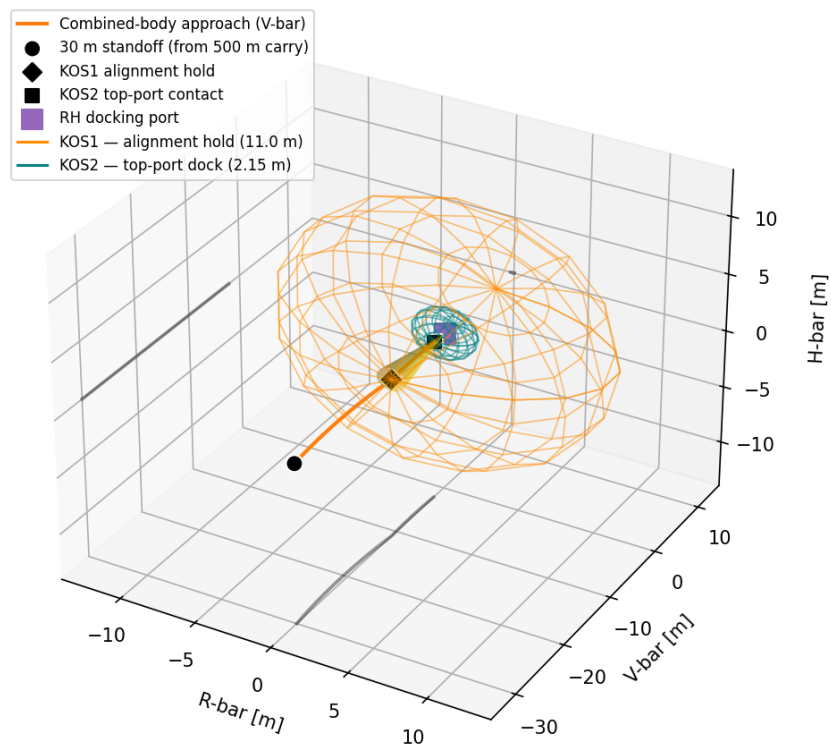


Fig 2b. Close approach + hard-dock: 30 m → KOS1_DOCK (11.0 m, clears the extended payload) → KOS2_DOCK (2.15 m, top-port contact). The off-axis payload forces the B2 re-point.

4. Results

ΔV is geometry-driven (mass-independent for translations); propellant is at each part's reference chaser mass. **Mission constants (per event):** DV_RPO_DEBRIS = 0.65, DV_RPO_DETUMBLE = 1.33, DV_RPO_TUG_MEET = 0.14, DV_RPO_RH_DOCK = 0.27, DV_RPO_RH = 0.41 m/s.

Table 1 — Part 1: Debris RPO (chaser 3228 kg, 6 °/s)

Phase	ΔV [m/s]	Prop [kg]	Time
P1 Approach (cut-off → V-bar hold)	0.051	0.066	86.7 min
P2 Inspection (V-bar hold: spin + axis)	0.005	0.006	8.0 h
P3 Transition to tumble axis (→ KOS1)	0.100	0.130	25.3 min
P4 Arm extension (deploy + lock onto tug)	0.011	0.015	1.5 min
P5 Spin-sync (match tumble)	0.168	0.219	0.4 min
P6a Final approach (KOS1 → KOS2)	0.050	0.065	3.2 min
P6b Momentum dump (detumble)	1.334	2.017	3.5 min
P7 Retreat (KOS2 → safe)	0.050	0.065	9.6 min
DV_RPO_DEBRIS (nominal ×1.50 abort)	0.653	0.850	—
DV_RPO_DETUMBLE (no abort margin)	1.334	2.017	—

Table 2 — Part 2 Phase A: Tug Meet (MS 1884 kg)

Phase	ΔV [m/s]	Prop [kg]	Time
A1 Close approach (30 m → KOS1)	0.050	0.038	13.1 min
A2 KOS1 hold (synch + comm)	0.000	0.000	30.0 min
A3 Capture run (KOS1 → KOS2 = arm)	0.050	0.038	3.6 min
A4 Arm extend + dock (CoM shift)	0.044	0.067	2.0 min
DV_RPO_TUG_MEET	0.145	0.143	—

Table 3 — Part 2 Phase B: RH Dock (MS 1884 kg + payload)

Phase	ΔV [m/s]	Prop [kg]	Time
B1 Carry (500 m hold → 30 m) — Fig 2c	0.054	0.099	313.3 min
B2 Re-point (combined inertia slew)	0.111	0.168	2.0 min
B3 Close approach (30 m → KOS1) — Fig 2b	0.050	0.091	12.6 min
B4 KOS1 hold (align + go/no-go)	0.000	0.001	30.0 min
B5 Hard-dock (KOS1 → KOS2, top port)	0.050	0.091	5.9 min
DV_RPO_RH_DOCK	0.266	0.449	—

5. Relevant Equations

- **Mean motion (CW):** $n = \sqrt{(\mu/a^3)}$ — at GEO $\approx 7.11 \times 10^{-5}$ rad/s.
- **Translation (2-impulse CW):** solve $\Phi_{rv} \cdot v_0^+ = r_f - \Phi_{rr} \cdot r_0$, then $\Delta V = |\Delta v_1| + |\Delta v_2|$, with $t = \text{chord}/(0.5 \cdot V_{MAX})$.
- **Stationkeeping:** $\Delta V = a_{dist} \cdot t$, with $a_{dist} = C_R \cdot (A/m) \cdot P_{sol} + a_{triax}$.
- **Attitude (detumble / spin-sync / re-point):** $H = |J \cdot \omega|$; $m_p = H/(\text{arm} \cdot v_e)$; $\Delta V = v_e \cdot \ln(m/(m - m_p))$; slew time = $H/(2 \cdot F_{edge} \cdot \text{arm})$; $\omega \approx V_{MAX}/r$.
- **Propellant (Tsiolkovsky):** $m_p = m_0 \cdot (1 - e^{-\Delta V/v_e})$, $v_e = I_{sp} \cdot g_0$.
- **Keep-out spheres:** KOS1 = $(\text{span}/2) \cdot 1.20$; KOS2(debris) = $\text{arm} + \text{tug}/2 = 5.60$ m; KOS2(tug-meet) = $\text{arm} = 5.00$ m; KOS2(dock) = $\text{MS}_h/2 + \text{port} = 2.15$ m; KOS1(dock) = $(\text{arm} + \text{tug} + \text{bus}) \cdot 1.20 \approx 11.0$ m.

Figures included: **Fig 1** (§3 Part 1, Table 1), **Fig 2a** (§3 Phase A, Table 2), **Fig 2c** and **Fig 2b** (§3 Phase B, Table 3).
Figures generated by `rpo_plots.py`; numbers by `rpo_run.py`.